

Example ^{A2} Two objects travel on the lines

$$\vec{r}_1(t) = \langle 2t+3, 4t+2, 3t+5 \rangle \quad -\infty < t < \infty$$

$$\vec{r}_2(s) = \langle s+2, 3s-1, -5s+10 \rangle \quad -\infty < s < \infty$$

will the objects' paths intersect?

will the objects collide with each other?

intersect: can we find t, s such that $\vec{r}_1(t) = \vec{r}_2(s)$?

collide: can we find t, s such that $\vec{r}_1(t) = \vec{r}_2(s)$ AND $t = s$?

if $\vec{r}_1(t) = \vec{r}_2(s)$ then

$$x: 2t+3 = s+2 \quad - \quad (1)$$

$$y: 4t+2 = 3s-1 \quad - \quad (2)$$

$$z: 3t+5 = -5s+10 \quad - \quad (3)$$

$$\text{from (1), } s = 2t+1$$

$$\text{Sub into (2) } 4t+2 = 3(2t+1)-1 \rightarrow \boxed{t=0 \text{ so, } s=1} \quad \text{check if these work in (3)}$$

$$(3): 3(0)+5 = -5(1)+10?$$

yes, so at $t=0, s=1$, $\vec{r}_1(t) = \vec{r}_2(s)$ they intersect

Q:- Find parametric eqs & symmetric eqs of line that passes through the pts A (2, 4, -3) & B (3, -1, 1). At what pts does this line intersect the xy-plane.

Sol $\vec{v} = \langle 3-2, -1-4, 1-(-3) \rangle = \langle 1, -5, 4 \rangle$

Direction numbers are $a = 1, b = -5, c = 4$. Taking pt (2, 4, -3) as P_0 , we see that parametric eqs are

$$x = 2 + t, \quad y = 4 - 5t, \quad z = -3 + 4t$$

Symmetric eqs

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{z+3}{4}$$

Line intersects xy-plane when $z = 0$. From parametric eq

$$z = -3 + 4t$$

$$0 = -3 + 4t$$

$$\boxed{t = \frac{3}{4}}$$

using value of t , we get $x = 2 + \frac{3}{4} = \frac{11}{4}$

$$y = 4 - 5\left(\frac{3}{4}\right) = \frac{1}{4}$$

Line intersects xy-plane at pt $\left(\frac{11}{4}, \frac{1}{4}, 0\right)$

Q:- Check if B1

$$L_1: x = 2 + t, \quad y = 21 + 7t, \quad z = 15 + 4t$$

$$L_2: x = -5 + 2s, \quad y = -18 + 9s, \quad z = -11 + 7s$$

intersect, skew or parallel. If they intersect determine pt of intersection

$$x = 2 + t, \quad y = 21 + 7t, \quad z = 15 + 4t$$

$$x = -5 + 2s, \quad y = -18 + 9s, \quad z = -11 + 7s$$

$$2 + t = -5 + 2s$$

$$21 + 7t = -18 + 9s$$

$$15 + 4t = -11 + 7s$$

$$t - 2s = -7$$

$$7t - 9s = -39$$

$$4t - 7s = -26$$

$$t = -3$$

$$s = +2$$

$$x = 2 - 3 = -1$$

$$y = 21 + 7(-3) = 0$$

$$z = 15 + 4(-3) = 3$$

Intersect at pt $(-1, 0, 3)$

Q7:- Find the pt at which the line ^{CT} ^{BT} pierces the plane

$$\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-4}{2}$$

pierces the plane $x+2y+2z=22$

Sol To find parametric eqs for the line, we introduce vector $\vec{r}$
Common ratio in the given symmetric eq $= \vec{n}$
the line
is

$$\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-4}{2} = t$$

which gives

$$x-2=t \quad y+3=2t \quad z-4=2t$$

$$x=2+t \quad y=-3+2t \quad z=4+2t$$

We want to find the values of 't' for which the variable pt
(x,y,z) on the line lies on the given plane. By substituting these eqs
into eq of plane we get

$$(2+t) + 2(-3+2t) + 2(4+2t) = 22$$

$$2+t-6+4t+8+4t=22$$

$$4+9t=22$$

$$9t=18$$

$$\boxed{t=2}$$

Desired pt is (4,1,8).

Q10:- A helicopter is to fly directly from a helipad at the origin in the direction of the pt $(1, 2, 1)$ at a speed of 30 ft/sec . What is the position of the helicopter after 20 sec ? Also find the distance covered?

Sol We place origin at the starting position of the helicopter. Then the unit vector

$$\vec{u} = \frac{1}{\sqrt{6}} \hat{i} + \frac{2}{\sqrt{6}} \hat{j} + \frac{1}{\sqrt{6}} \hat{k}$$

gives the flight direction of the helicopter. The position of the helicopter at any time t is

$$r(t) = r_0 + t (\text{Speed}) \vec{u}$$

$$r(t) = 0 + t (30) \left(\frac{1}{\sqrt{6}} \hat{i} + \frac{2}{\sqrt{6}} \hat{j} + \frac{1}{\sqrt{6}} \hat{k} \right)$$

$$r(t) = \frac{30}{\sqrt{6}} t (\hat{i} + 2\hat{j} + \hat{k})$$

When $t = 20 \text{ sec}$

$$r(20) = \frac{600}{\sqrt{6}} (\hat{i} + 2\hat{j} + \hat{k})$$

$$r(20) = \left\langle \frac{600}{\sqrt{6}}, \frac{1200}{\sqrt{6}}, \frac{600}{\sqrt{6}} \right\rangle$$

After 20 sec of flight from the origin towards $(1, 2, 1)$, the helicopter is located at the pt $\left(\frac{600}{\sqrt{6}}, \frac{1200}{\sqrt{6}}, \frac{600}{\sqrt{6}} \right)$ in space. It has covered a distance of $(30 \text{ ft/sec})(20 \text{ sec}) = 600 \text{ ft}$, which is length of $r(20)$